Advertising Sales Prediction Analysis

Understanding the Impact of TV, Radio, and Newspaper Advertising on Product Sales

Dataset Overview

About the Dataset

The dataset contains advertising expenditure data across three key platforms and their corresponding sales performance.

Dataset Size: 200 records with 4 variables

Variables:

TV
Television advertising spend

Radio
Radio advertising spend

Newspaper
Newspaper advertising spend

Sales
Product units sold

Question 1: Average TV Advertising Spend

$147.04

Average TV Advertising Expenditure

💡 Insight: TV advertising represents a significant investment for companies in this dataset, averaging nearly $150 per campaign.

Question 2: Radio Advertising & Sales Correlation

Correlation Coefficient

0.35

Moderate Positive Relationship

What This Means

- Radio advertising shows a **moderate positive correlation** with sales
- As radio spending increases, sales tend to increase moderately
- The relationship is weaker compared to other advertising channels

Key Takeaway

While radio advertising does contribute to sales growth, its impact is moderate. Companies should consider radio as part of a broader multi-channel strategy rather than relying on it as a primary sales driver.

Question 3: Which Medium Has Highest Impact?

📺 **TV Advertising**

Correlation with Sales: 0.90

1st	2nd	3rd
TV	**Radio**	**Newspaper**
0.90	**0.35**	**0.16**
correlation	correlation	correlation

🎯 TV advertising shows the strongest correlation (0.90) with sales, making it the most impactful medium for driving product sales in this dataset.

Question 4: Multiple Linear Regression Model

Model Specification

Predictors:
- TV Advertising
- Radio Advertising
- Newspaper Advertising

Target Variable:
- Sales (units)

Model Performance

The scatter plot shows actual sales vs. predicted sales, with points closely following the diagonal reference line. This indicates strong model accuracy with most predictions aligning well with actual values.

Key Findings from the Model

Strong Predictive Power: The model successfully captures the relationship between advertising spend and sales

Linear Relationship: The tight clustering around the diagonal line confirms a linear relationship between predictors and sales

Minimal Outliers: Few data points deviate significantly from predictions, suggesting consistent advertising effects

Question 5: Predicting Sales for New Campaign

Campaign Budget Allocation

TV	$200	67%
Radio	$40	13%
Newspaper	$50	17%
Total Budget	**$290**	

Predicted Sales

19.87

units

💡 Strategic Insight: With this budget allocation heavily weighted toward TV ($200), the model predicts approximately 20 units in sales. The high TV investment aligns with its strong correlation to sales performance.

Question 6: Impact of Data Normalization

Metric	Without Normalization	With Normalization
R^2 Score	0.9059	0.9059
RMSE	1.785	1.785
Difference	—	Negligible

Conclusion

Normalization has minimal impact on model performance. The R^2 scores and RMSE values remain virtually identical, indicating that the original feature scales were already well-suited for the linear regression model. This suggests the data is naturally well-conditioned for analysis.

Question 7: Excluding TV Advertising

What happens when we remove the most impactful predictor?

All Variables Model

TV + Radio + Newspaper

R^2 Score

0.90

RMSE: 1.70

Radio + Newspaper Only

Excluding TV

R^2 Score

0.11

RMSE: 5.24

⚠️ **Critical Finding: Removing TV advertising causes model performance to collapse. R^2 drops 89% and prediction error triples, confirming TV is the dominant sales driver.**

Key Insights & Recommendations

TV is King

TV advertising shows the strongest correlation (0.90) with sales and is essential for accurate predictions. Removing it causes an 89% drop in model accuracy.

Multi-Channel Approach

While TV dominates, radio (0.35) and newspaper (0.16) provide complementary value. The best model uses all three channels together.

Linear Relationship

Sales respond linearly to advertising spend, making predictions reliable and budgets scalable. Each dollar invested yields consistent returns.

Recommendation: Allocate 60-70% of advertising budgets to TV for maximum impact, with radio and newspaper supporting specific demographic or regional targets.

Thank You

Questions & Discussion

📊 Advertising Sales Analysis 2026
Made By:- Om A Channawar